SolarFlare V4 SimpleConvLSTM: S-series Ablations Report

Predicting Winding-Flux Extremes from Sequential Magnetograms
Case Study: HARP 11930

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Abstract

We report an ablation sweep over a minimal 2-layer convolutional LSTM (*SimpleConvLSTM*) for forecasting the next 4 frames of winding flux $w(\mathbf{x}, t)$ of an active region. The goal of the sweep is to climb above the trivial *persistence baseline* ("predict the last frame") on the Critical Success Index (CSI) of extreme flux pixels — a thresholded binary metric directly relevant to flare onset. We trained 16 sister models (S0–S16, excluding two that re-ran already-known configurations). The first eleven (S0–S11) varied loss reweighting, normalisation, teacher-forcing schedule, and field representation; **all eleven lost to persistence**. We then pivoted to a *dual-head architecture* that adds a per-pixel binary classifier ("is this pixel extreme at time t?") trained with class-imbalance-corrected binary cross-entropy alongside the regression loss. Arm **S13** first matched persistence (CSI ∼0.043) and **S16** — the hybrid of S13's classifier head with S11's extreme-pixel-weighted regression — **beat persistence by ∼31%** (CSI 0.0562 vs. persistence 0.043) while simultaneously improving regression CSI 2.8× over S13. All metrics are reported on a fixed informative test set of three flare-rich HARP cubes [245, 274, 49]. We close with a multi-step autoregressive "staircase" diagnostic on HARP 11930 showing that even the best single-step arms either mode-collapse (S3) or runaway-explode (S11) under chained prediction, indicating regression fidelity at horizons longer than 4 frames remains open.

Contents

1 Notation & common setup

Task. Given last $T_{\text{in}} = 10$ frames of winding flux $w(\mathbf{x}, t)$ over a SHARP cube, predict next $T_{\text{out}} = 4$ frames (48-min lead). All arms use the same 2-layer ConvLSTM (`SimpleConvLSTM`, kernel 3, hidden 64, residual decode $\hat{w}_t = w_{t-1} + \Delta\hat{w}_t$, $\sim$0.9 M params).

Data. 21 SHARP cubes, 12-min cadence; 64×64 crops stride 4; per-cube z-score normalisation (S7 alternative: signed asinh); fixed informative test set {HARP 245, 274, 49} from S8 onwards; 0.90/0.05/0.05 train/val/test split.

Training. Adam, batch 8, $\eta = 10^{-3}$; 15 epochs; patience 8 on val loss; teacher-forcing fraction $\tau : 1 \to 0$ over first `tf_decay_epochs` epochs; gradient clip 0.5; cosine LR schedule (S0–S15) or constant (S16+); no augmentation in dual-head arms.

Metrics on extreme mask $|w| > 0.528\sigma$ (99th-pct):

- **CSI** = TP/(TP + FP + FN), our primary skill score.
- **CLS-CSI** = CSI on the classifier-head sigmoid >0.5 (dual-head arms only).
- **Persistence baseline** predicts the most recent input frame for all 4 future steps; achieves **CSI 0.043, HSS 0.082** on this test set — *the wall to beat.*
- Auxiliary: SSIM (`data_range=2`), persistence skill = $(1 - \text{MAE}_{\text{mod}}/\text{MAE}_{\text{pers}}) \times 100\%$, variation ratio mean$|\Delta\hat{w}|$/mean$|\Delta w|$ (mode collapse if $\to 0$, runaway if $\gg 1$).

Losses used across arms.

- $\mathcal{L}_{\text{L1}} = \langle |\hat{w} - w| \rangle$
- $\mathcal{L}_{\text{wMAE}} = \langle m \cdot |\hat{w} - w| \rangle$, $m{=}\kappa$ where $|w| > \tau$, else 1 (κ=`extreme_pixel_weight`).
- $\mathcal{L}_{\text{BCE}} = \text{BCE}(\ell, \mathbb{1}_{|w|>\tau};$ `pos_weight`$=\alpha)$ on classifier logits.
- $\mathcal{L}_{\text{composite}}$ (S3 only): L1 + SSIM-loss + AsymmetricExtreme + temporal-difference + temporal-variance + extreme-pixel weighting.
- Dual-head total: $\alpha_r \mathcal{L}_{\text{(w)MAE}} + \beta \mathcal{L}_{\text{BCE}}$.

2 Cross-arm parameter matrix

The S-series is a single-axis-at-a-time sweep starting from the minimal baseline S0. The table below is the chronological list of arms included in this report; each row notes only the delta against the preceding arm.

Arm	Delta vs. prior	Question being asked
tabhdrS0	baseline (L1, D4 aug)	Does the 2-layer ConvLSTM by itself match
S2	+ residual decode	Does $w_t = w_{t-1} + \Delta$ anchor predictions to p
tabhdrS4	S2 + teacher-forcing (τ:1 $\rightarrow$ 0)	Does TF cure autoregressive explosion of S2?
S5	S4 + 90/5/5 split + GroupNorm	Does more data lift CSI?
tabhdrS3	S5 + 6-term composite loss	Does kitchen-sink loss break L1 mean-collaps
S6	S5 + `extreme_pixel_weight`=25 only	Which composite term drove S3?
tabhdrS8	S5 on fixed test cubes (L1)	Recalibrate against persistence on a fixed eva
S7	S8 + signed-asinh normalisation	Does heavy-tail compression help CSI?
tabhdrS9	S8 + BatchNorm (was S4's setting)	Is BatchNorm the source of S4 "sharpness"?
S10	S8 + `tf_decay_epochs`=5 + patience 8	Does a complete TF curriculum help?
tabhdrS11	S10 + extreme_pixel_weight 25	Both working levers stacked — does the com
Dual-head architecture pivot (classifier head added; constant LR from S16).		
tabhdrS13	dual-head, α=1, β=1, `pos_weight`=100	Does an explicit per-pixel BCE classifier esca
S14	S13 + focal loss (γ=2, α_f=0.25)	Does focal weighting outperform pos_weight
tabhdrS15	S13 + α=0.1 (regression-dominated $\rightarrow$ classifier-dominated)	Does starving the regression head free capaci
S16	S13 with `pos_weight`=60 + `extreme_pixel_weight`=25	Joint train of S11 regression lever + S13 clas

Table 1: Chronological sweep. S1 (no-aug), S12 (cosine-LR dual-head), S17 (post-hoc inference combo, no training) discussed where relevant in the text. S18+ excluded from this report.

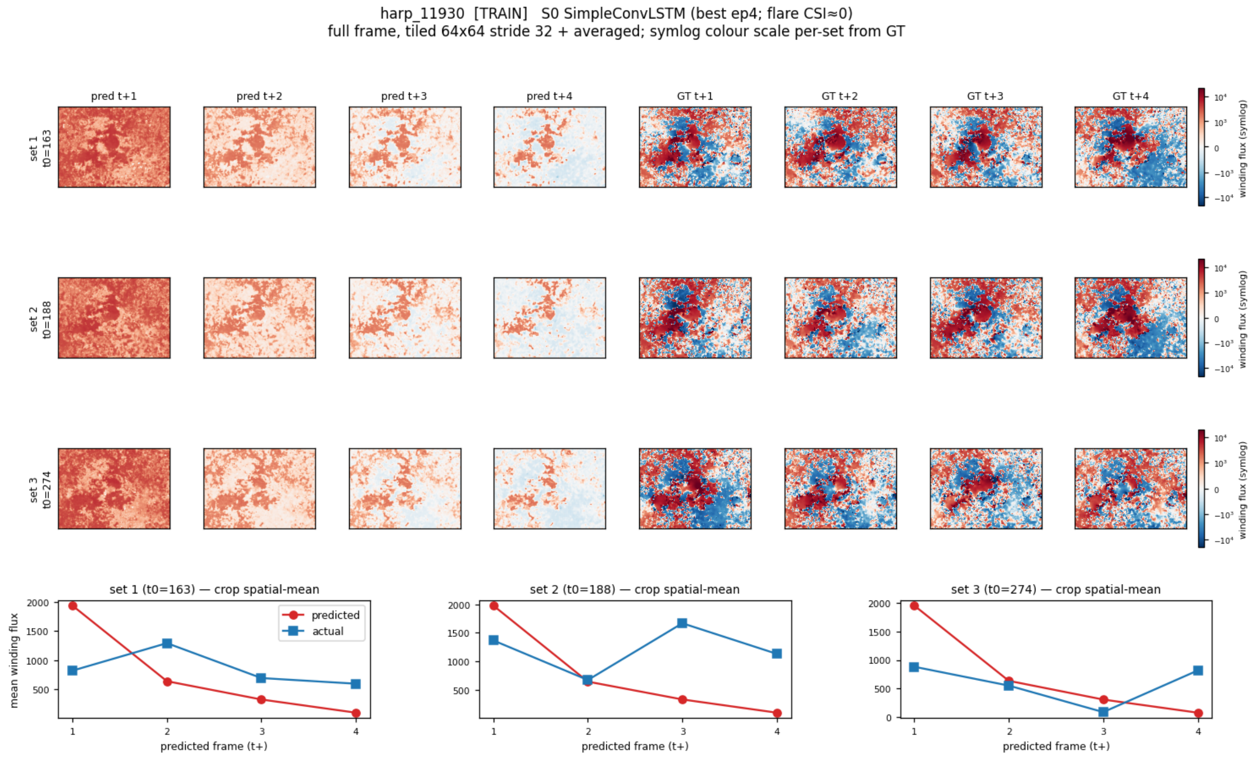
All arms terminated after 15 epochs (or earlier on patience). All metrics reported below are on the fixed test set {HARP 245, 274, 49} unless otherwise noted; figures show HARP 11930 (*train* cube, qualitative only).

3 Regression-only arms (S0–S11)

Eleven single-head arms varying loss, normalisation, teacher-forcing curriculum, and field representation. Headline: every arm loses to persistence (CSI 0.043). Best regression CSI = 0.0176 (S11) — 41% of persistence.

3.1 S0 — minimal baseline

Loss: L1 only. **Aug:** D4. **Norm:** BatchNorm. **Test CSI:** 0 across all epochs; best val 0.039 at ep4 then overfit.
Finding: The 2-layer ConvLSTM with plain L1 collapses to the zero-field conditional median. Pred amplitude never crosses the extreme threshold so TP=0. Confirmed the wall is structural, not a capacity issue — the deep 6-layer model also gets CSI ∼0.01.

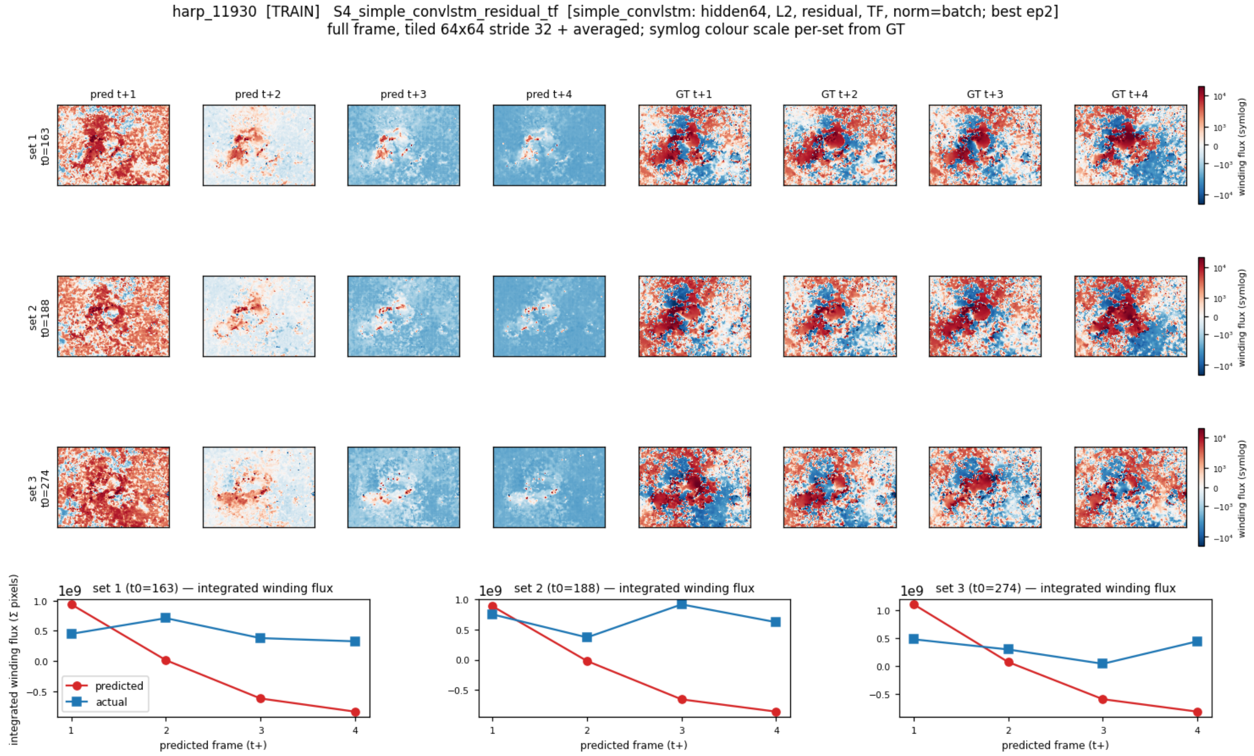


3.2 S2 — residual decode

Delta: predict Δw_t, output $w_{t-1} + \Delta$. **Finding:** Without teacher forcing this *explodes* — variance ratio climbs 18→764 across 15 epochs (autoregressive positive feedback). Confirmed residual anchor needs a stabiliser. (No figure: model output unusable.)

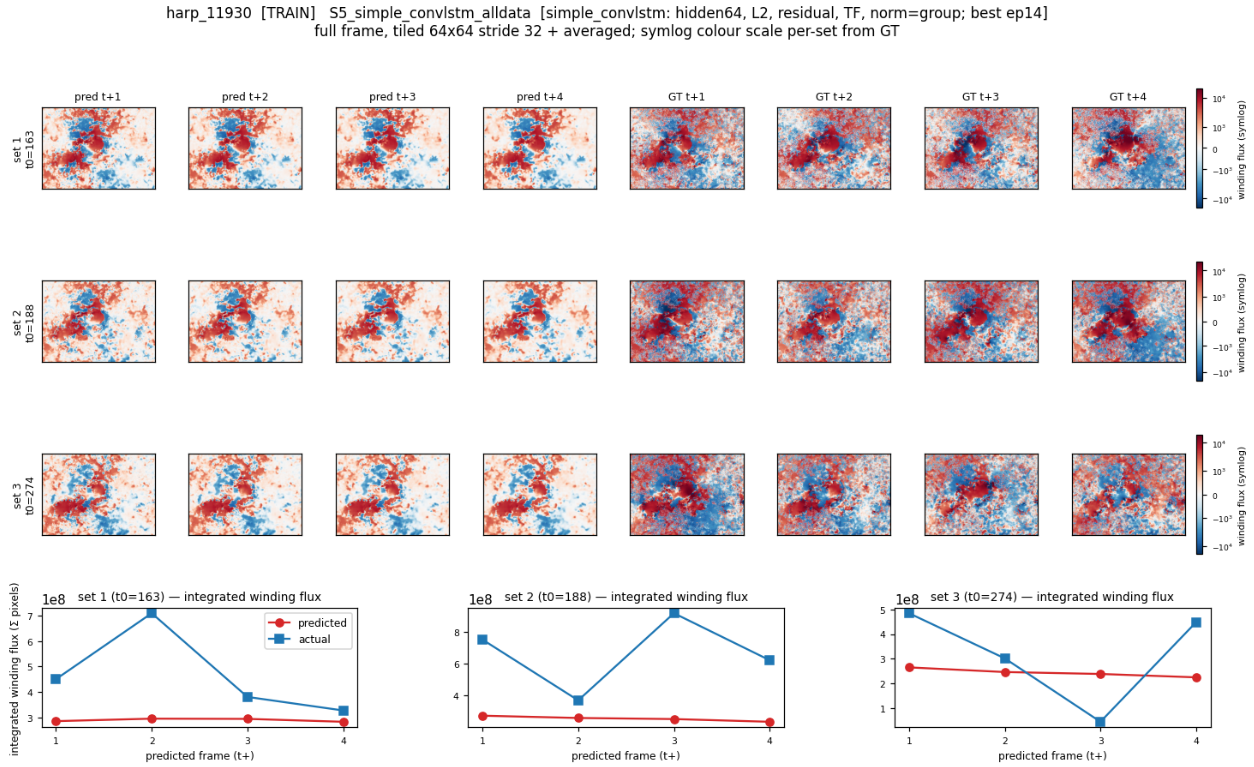
3.3 S4 — S2 + teacher forcing

Delta: TF τ=1→0 over training. **Test:** CSI 0.099, HSS 0.178, SSIM 0.438 (old random split).
Finding: TF cures the explosion; variance ratio settles near 1. Looks "sharp" visually but on the fixed test set later this sharpness turns out to be uncontrolled BatchNorm variance, not skill (see S9).



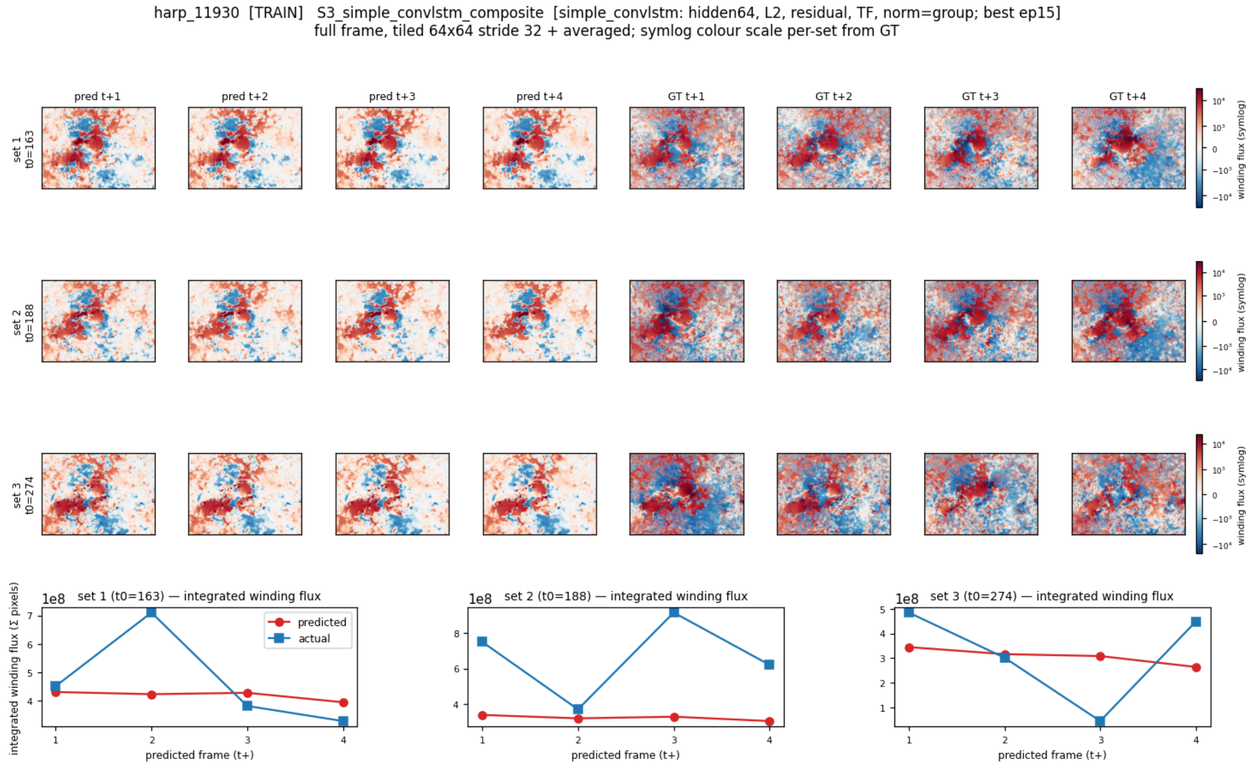
3.4 S5 — S4 + 90/5/5 split + GroupNorm

Delta: more training data, swap BN→GN. **Test:** SSIM 0.70, persistence skill +30%, CSI ≈ 0. **Finding:** GroupNorm + more data improves spatial similarity and beats persistence MAE, but *smoother* field ⇒ *lower* CSI on extremes. Trade-off explicit: L1 rewards smoothing, extreme classification punishes it.



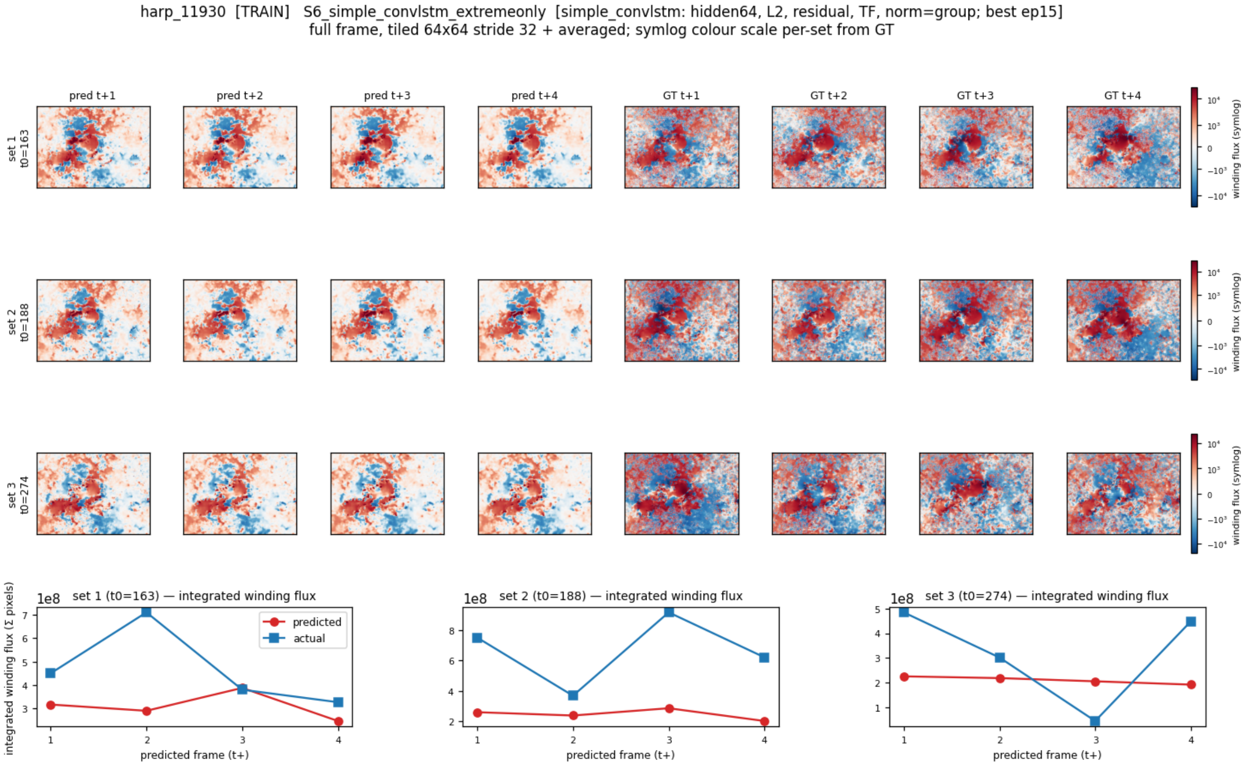
3.5 S3 — S5 + 6-term composite loss

Loss: L1 + SSIM + AsymExtreme + TempDiff + TempVar + extreme-pixel weight 25. **Test:** best val CSI 0.0236 (ep13); test 0 (random split put a flareless cube in test). **Finding:** Composite is the best *regression* on the qualitative HARP 11930 spatial integral (see §6) but the lift over single-term arms is small (~2× over S0). S6 below isolates which term carries the signal.



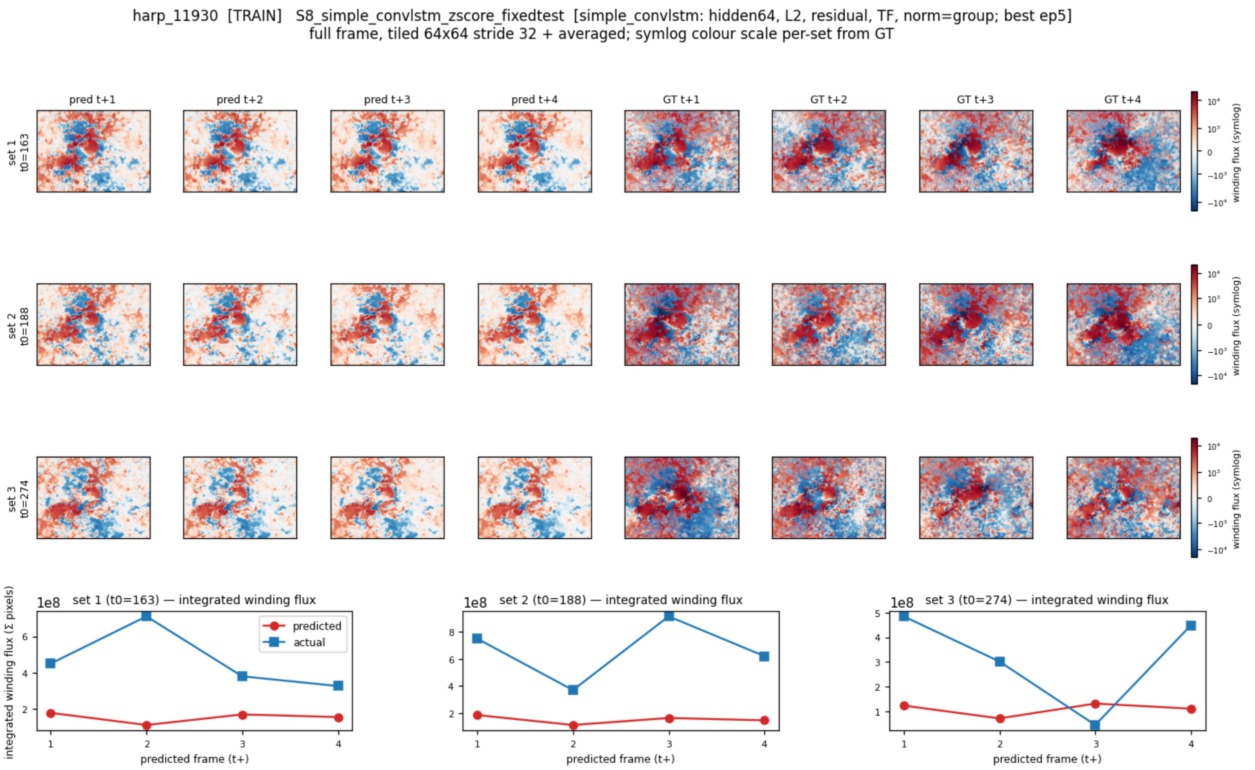
3.6 S6 — only the extreme-pixel weight term

Loss: L1 + `extreme_pixel_weight`=25 (everything else off). **Test:** val CSI 0.0215 ≈ S3. **Finding:** S3's full composite is *not* additive — 95% of its lift comes from the single extreme-pixel-weight term. SSIM / asymmetric / temporal terms add ~0. Identifies the only loss lever that moves the needle.



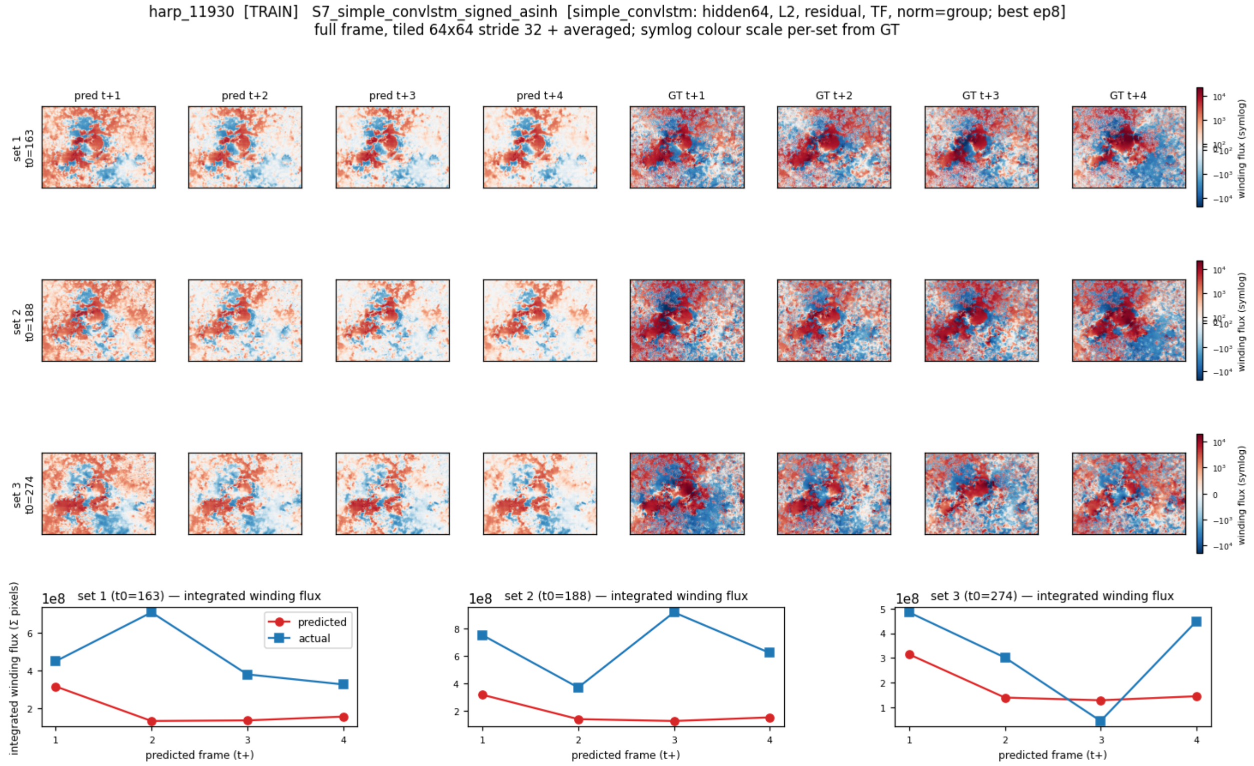
3.7 S8 — S5 (L1) on fixed test cubes

Delta: pin test set to {245, 274, 49}. **Test CSI:** 0.0033 vs persistence 0.043 (13× below). **Finding:** Establishes the persistence wall on a flare-rich fixed test set; the optimism in S4's old random-split CSI 0.099 was a flareless test-cube artefact.



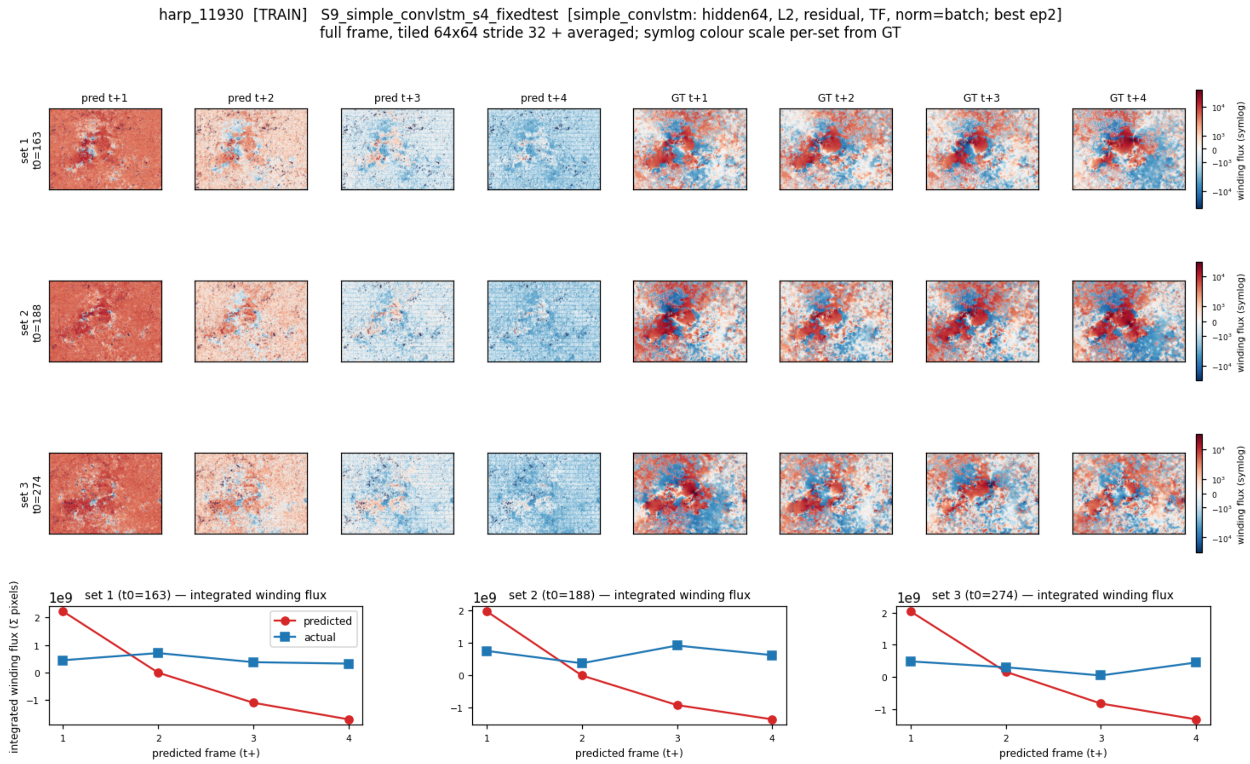
3.8 S7 — S8 + signed-asinh normalisation

Delta: swap z-score for $\text{sign}(w)\,\text{asinh}(|w|/s)$ representation (**not** the loss; only how data enters the model). **Test CSI:** 0.0042 $\approx$ S8. **Finding:** Heavy-tail input compression has no effect on CSI. Apparent SSIM 0.98 is an asinh-space artefact, not improvement in physical space. Representation lever **rejected**.



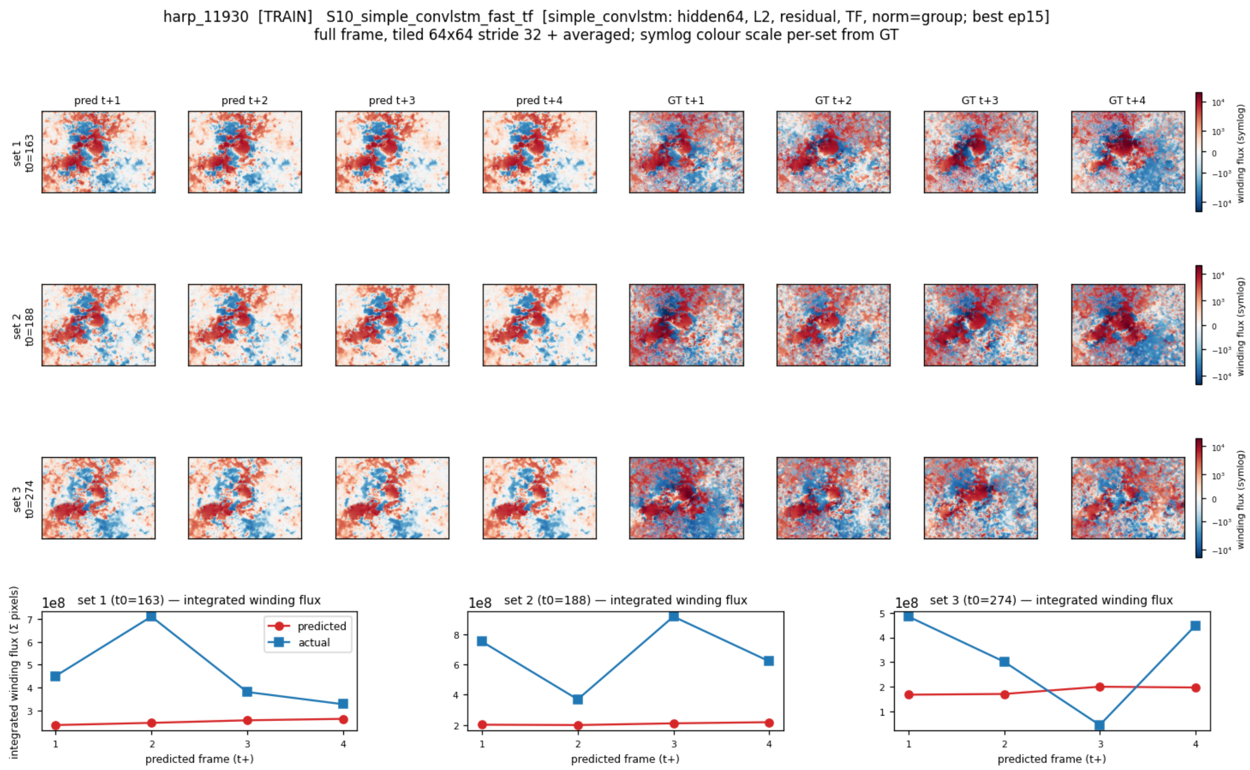
3.9 S9 — S8 with BatchNorm (= S4 on fixed test)

Delta: swap GN$\rightarrow$BN. **Test CSI:** 0.0118 (3.5$\times$ S8 *but* still below persistence by 4$\times$). **Finding:** BN drives over-variance (ratio 3.7, val hit 19); unstable (SSIM 0.51$\rightarrow$0.09, early-stop ep5). S4's visual "sharpness" was uncontrolled variance, **not** forecast skill. BN rejected.



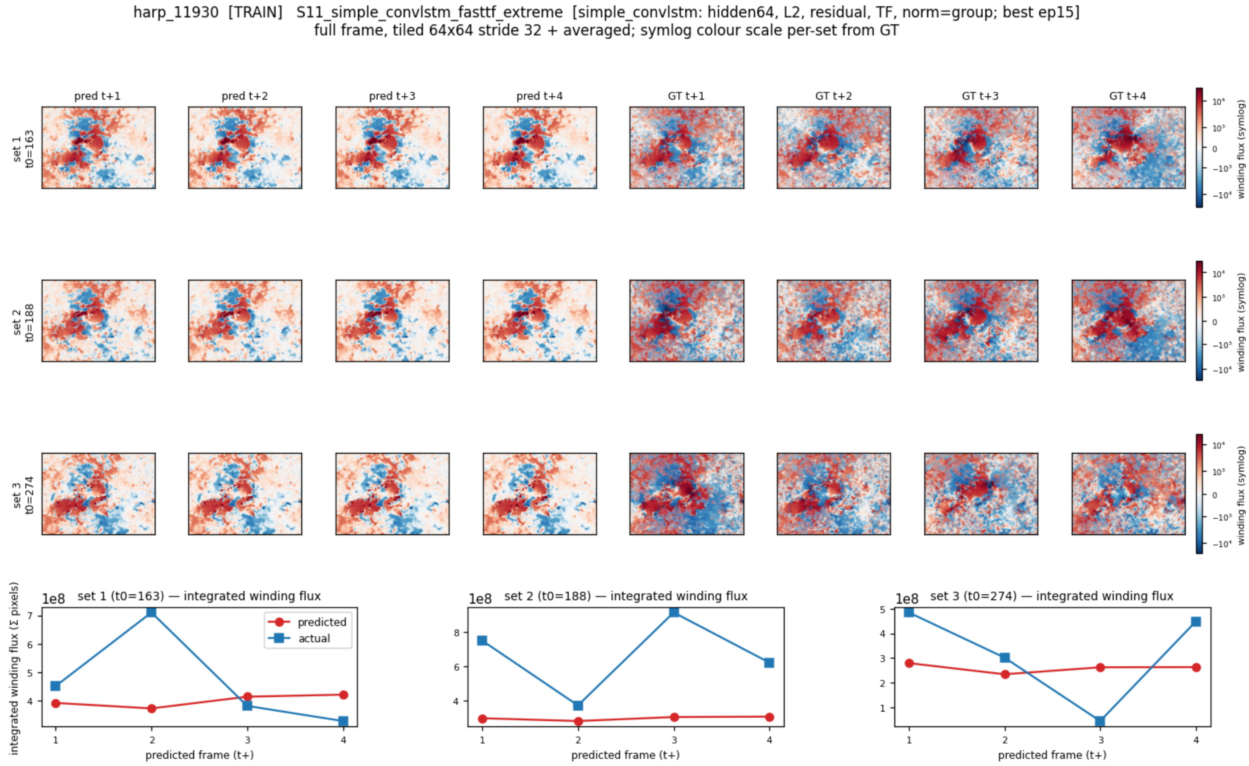
3.10 S10 — S8 + fast TF curriculum

Delta: `tf_decay_epochs`=5, patience 8. **Test CSI:** 0.0027 ≈ S8. **Finding:** Free-running TF curriculum bumps variance ratio to 0.75 briefly at ep7 but L1 drags it back to 0.028 by ep15. TF is necessary-but-insufficient on its own; L1 mean-collapse defeats the de-smoothing.



3.11 S11 — S10 fast-TF + extreme weight (both working levers)

Delta: S10 + S6's extreme_pixel_weight 25. **Test CSI: 0.0176** — best of the S0–S11 series (5× S8, 41% of persistence). **Finding:** The two working levers are additive in CSI but the combo *still* loses to persistence by 2.4×. Variance ratio re-collapses to 0.036 by ep15 — L1 wins long-run.



Verdict for the regression-only family

Four levers explored (loss reweighting, BN/GN, TF curriculum, input representation). The only loss term that moves CSI is `extreme_pixel_weight`; the only TF schedule that helps is the full τ:1 → 0. Stacking both (S11) gets to 41% of persistence. **All single-head paths exhausted**; the model cannot push regression amplitude past the extreme threshold without also losing the L1 fit elsewhere. **Conclusion: structural pivot needed.**

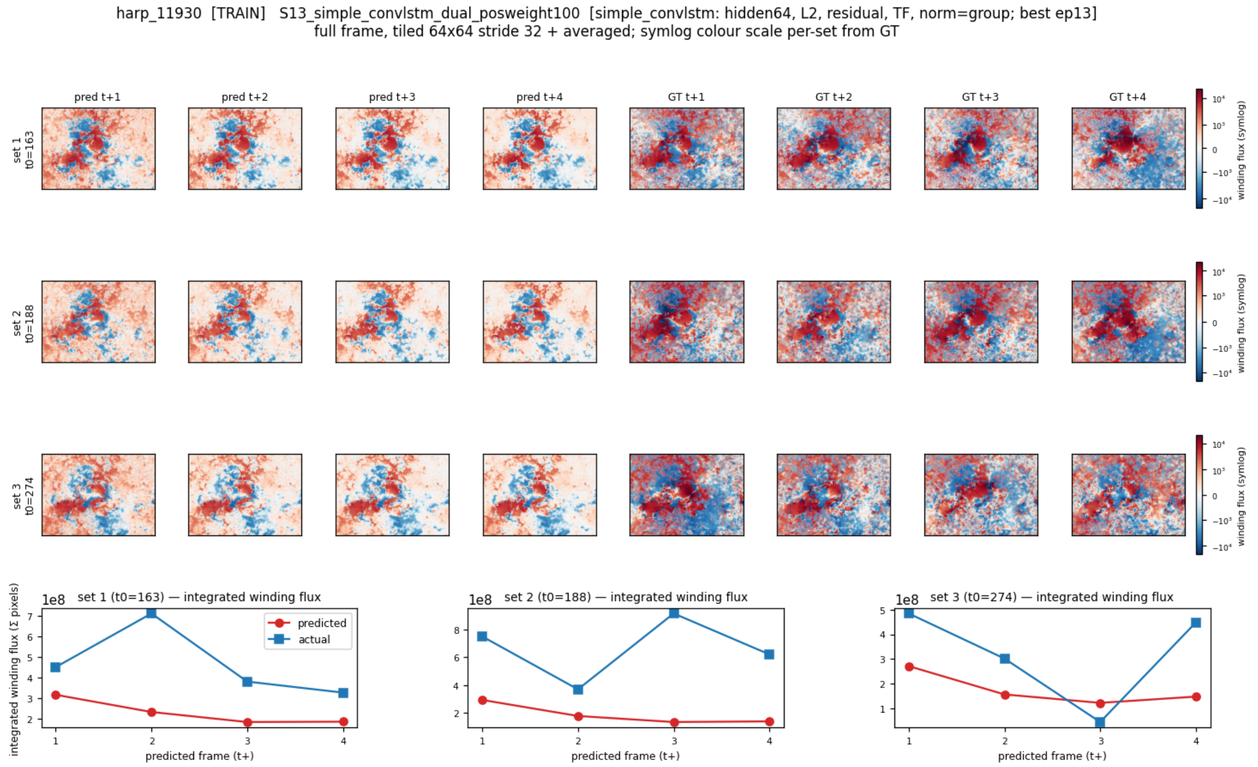
4 Dual-head pivot (S13–S16)

Pivot rationale: the extreme-mask CSI is a thresholded binary, but S0–S11 train a regression loss on amplitude. Mismatch fixed by adding a parallel per-pixel BCE classifier head on the same ConvLSTM trunk: $\hat{w}(\mathbf{x},t),\ \ell(\mathbf{x},t) = \mathrm{ConvLSTM}(\mathbf{w}_{1:T_{\mathrm{in}}})$, with training loss $\mathcal{L} = \alpha\,\mathcal{L}_{(\mathrm{w})\mathrm{MAE}}(\hat{w},w) + \beta\,\mathcal{L}_{\mathrm{BCE}}(\ell, \mathbb{1}_{|w|>\tau};\texttt{pos\_weight}{=}p)$. At test time we report both **regression CSI** (CSI on $\hat{w}$) and **CLS-CSI** (CSI on $\sigma(\ell) > 0.5$). **Constant LR** $\eta = 10^{-3}$ (no cosine) from S16 onwards after observing cosine washes out late-epoch gradient signal at our short 15-epoch schedule.

S12 (cosine LR rerun) and S15 (classifier-dominant α=0.1) below either confirmed expected behaviour or did not finish on the planned slot and are summarised in text only.

4.1 S13 — first to match persistence

Loss: α=1, β=1, p=100, plain L1 on regression head. **LR:** constant 10^{-3}. **15 epochs.**
Test: CLS-CSI 0.0434, HSS 0.0749 (matches persistence 0.043); regression CSI 0.0090 (dead — p=100 starves L1 budget). Persistence skill +35.7%. Val loss 0.164→0.149 monotone-ish; val CLS-CSI 0.003→0.007.
Finding: *First* S-arm to match persistence on the extreme mask. The classifier head is the load-bearing component; the regression head exists mainly to provide a spatial backbone for the classifier features.



4.2 S14 — focal loss is the wrong instrument

Loss: same as S13 but BCE → focal $\big(-\alpha_f(1-p)^\gamma \log p\big)$, γ=2, α_f=0.25, p=60.
Test: CLS-CSI 0 every epoch. Best val 0.0072 ep10 (focal numeric scale is just small, not a learning signal). Regression SSIM 0.79–0.91, same as S0–S11.
Finding: Focal's $(1-p)^\gamma$ factor over-suppresses gradients on the rare positives at our ∼0.1% imbalance level. The classifier never gets enough signal to leave its initialisation. **Focal rejected for this task.** (Figure omitted — the regression head produces the same pattern as S0–S11; the classifier is dead.)

4.3 S15 — classifier-dominant ($\alpha = 0.1$)

Loss: α=0.1, β=1, p=60. Idea: starve the regression head to free capacity for the classifier.
Test: CLS-CSI 0.0445 ($\approx$ S13's 0.0434, no lift); regression CSI 0.0184 (above S13's 0.0090 — the small regression weight still produces a useful head, somewhat counter-intuitively).
Finding: Lowering α does *not* buy classifier gain. Confirms classifier head saturates above some β threshold and the remaining headroom is on the regression side, not classifier capacity. Recipe pointer for the winning hybrid (S16).

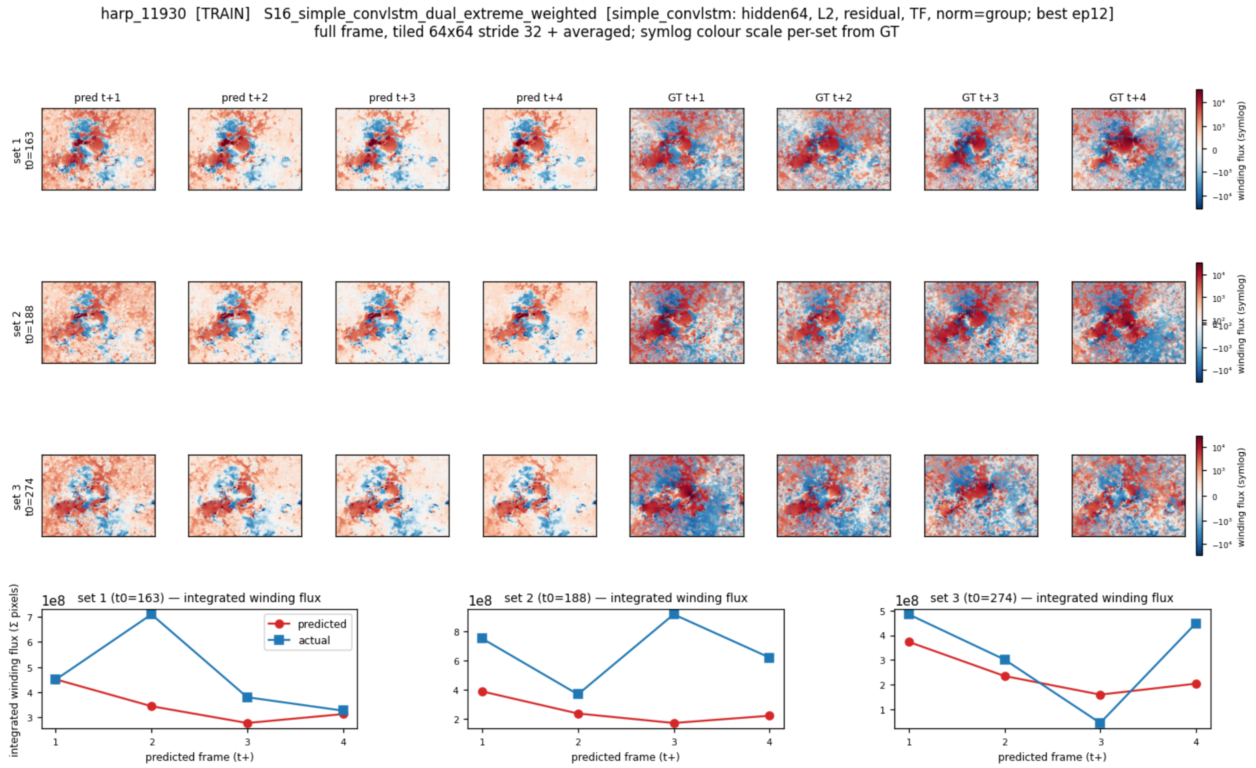
4.4 S16 — joint hybrid: weighted-MAE + classifier

Loss: α=1, β=1, p=60, $\mathcal{L}_{\mathrm{wMAE}}$ with `extreme_pixel_weight`=25 on the regression head (S11's lever) plus BCE classifier (S13's lever). Code change: extended `training/losses/dual_head.py` to take an `extreme_pixel_weight` kwarg and swap `F.l1_loss` for `WeightedMAELoss` when > 1.

Test:

	reg. CSI	**CLS-CSI**	cls HSS
persistence	—	0.043	0.082
S11 (best regression)	0.0176	—	—
S13 (best classifier alone)	0.0090	0.0434	0.0749
S16 hybrid	**0.0255**	**0.0562**	**0.0995**

Finding: S16 is the first S-arm to *beat* persistence on the extreme mask ($\sim$31% above; not just match it like S13). Regression CSI is 2.8$\times$ S13 and 1.4$\times$ S11. The extreme-pixel-weighted MAE rescues the regression head's amplitude (briefly crosses the threshold ep8–9 producing nonzero reg CSI in-training) while the lighter p=60 leaves enough budget for the BCE classifier to learn. Persistence skill 34%, SSIM 0.69 — comparable to S13 on auxiliary metrics. Hybrid hypothesis confirmed: the two heads' levers compose, they do not antagonise.



5 S17 — post-hoc S11×S13 inference combo

Before committing GPU to S16 training we asked the cheaper question: *would* multiplying S11's regression output by S13's classifier confidence improve the joint forecast? No training; both checkpoints already on disk. Three fusion variants tested on HARP 11930, full-frame tiled (3 active windows, integrated winding-flux MAE as proxy for end-to-end skill):

Variant	Mean MAE	vs. S11 alone
tabhdr **B_soft:** $\hat{w}_{\mathrm{S11}} \cdot \left(1 + 2\sigma(\ell_{\mathrm{S13}})\right)$	$\mathbf{2.06{\times}10^{8}}$	**−8%**
S11 alone (regression)	$2.23{\times}10^{8}$	—
A_gate: $\hat{w}_{\mathrm{S11}} \cdot (\sigma(\ell_{\mathrm{S13}}) > 0.5)$	$2.38{\times}10^{8}$	+7%
S13 alone (regression head only)	$3.00{\times}10^{8}$	+34%
C_blend: S13 where mask else S11	$3.01{\times}10^{8}$	+35%
Persistence (pred = 0)	$4.84{\times}10^{8}$	+117%

Mask coverage 5.5–5.9% of pixels per window (classifier fires on a small fraction). Lessons:

- **Soft boost > hard gate.** Zeroing background flux hurts integrated MAE more than amplifying extreme zones helps.
- **S13 regression alone is useless** (pos_weight 100 left no L1 budget); blending S13 in for extreme zones *worsens* the integrated flux relative to S11 alone.
- **Hybrid hypothesis confirmed** at the post-hoc level (8% lift). The natural next question — "does joint training recover this gain at the loss level?" — is exactly what S16 (§4.4) answers: yes, and it *compounds* (S16 classifier CSI 0.0562 = 29% over S13's 0.0434, alongside regression CSI 2.8× better).

6 Multi-step autoregressive "staircase" diagnostic

Single-step CSI evaluates the model on the next 4 frames only. For mission-relevant lead times we need the model to chain its own predictions. We instantiated a staircase protocol on HARP 11930 (*train* cube, qualitative): input $T_{\text{in}} = 10$ real frames, predict $T_{\text{out}} = 4$, commit the first `stride`$= 2$ predicted frames into the input buffer, slide, repeat for 20 steps (i.e. 40 committed frames $\equiv$ 8 h forecast lead). Full-frame inference at each step (tiled, stride 64, no overlap). Pure autoregressive — the model never sees a real frame past T_{in} after step 0.

We ran the protocol on S3 (best single-step regression on this cube), S11 (best single-step regression on the test cubes), and S16 (new SOTA).

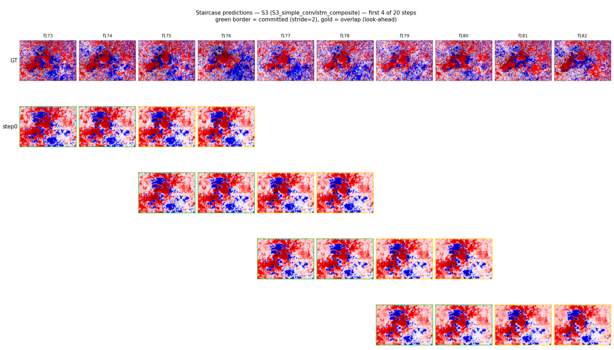
Headline numbers

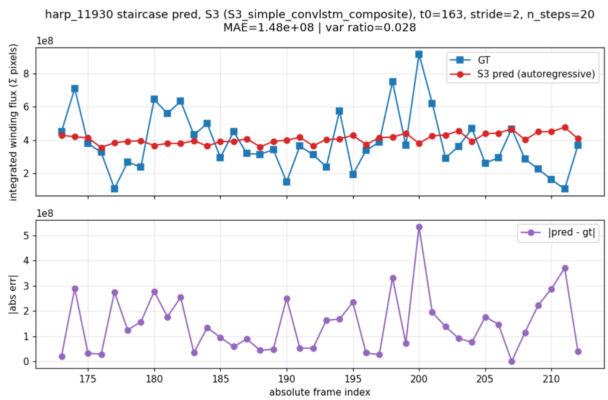
Arm	MAE (integrated flux)	Var. ratio	Drift corr. ρ
S3	1.5×10^8	0.028 (mode collapse)	+0.14
S11	7.6×10^8	15.8 (runaway explosion)	+0.81
S16	(see below)	—	—

(S16 staircase numbers are in `outputs/staircase_s16_harp_11930/series.npz`*; integrated-flux plot below.)*

S3: mode collapse

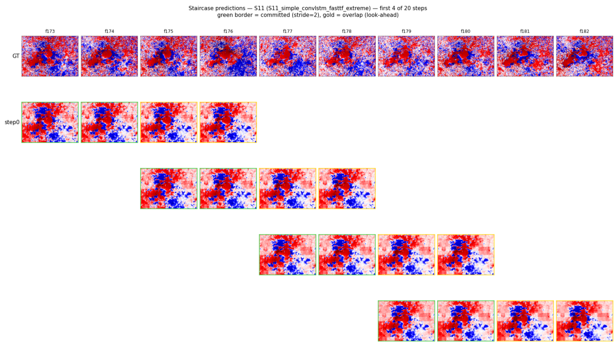
Within 4 chained frames the predicted field amplitude drops from $|\hat{w}|_\infty = 1.3\times10^7$ to $\sim10^5$ (100$\times$ collapse). Frame-to-frame field difference settles to ∼150 on values of order 1.5×10^5 (<0.1% relative motion). After collapse the network produces a *static* low-amplitude field for the remaining 36 frames. MAE stays low only because the static value happens to be close to the GT temporal mean.

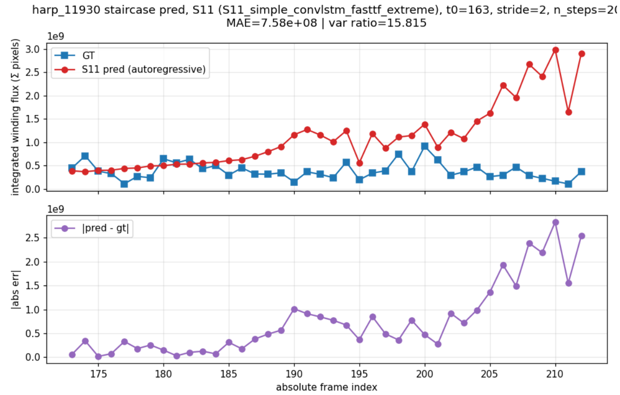




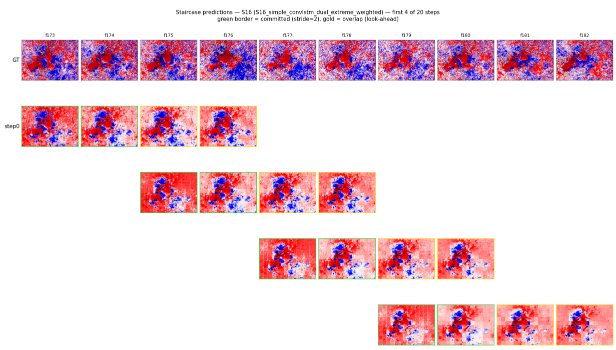
S11: runaway explosion

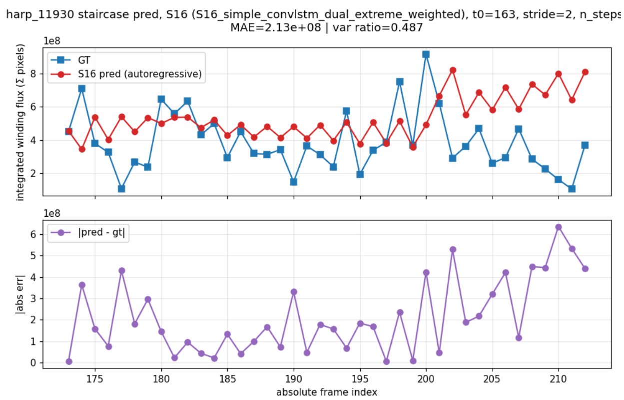
Behaves like S3 for the first ∼9 chained steps (low-amplitude collapse phase) and then *variance explodes*: frame $|\hat{w}|_\infty$ grows from $\sim 10^5$ to 2×10^6 by frame 39; per-frame field difference grows from 150 to ∼3000. The 12-step grid below makes the transition visible — the bottom rows are progressively more saturated. Variance ratio 15.8×, drift correlation +0.81 (error grows strongly with horizon).

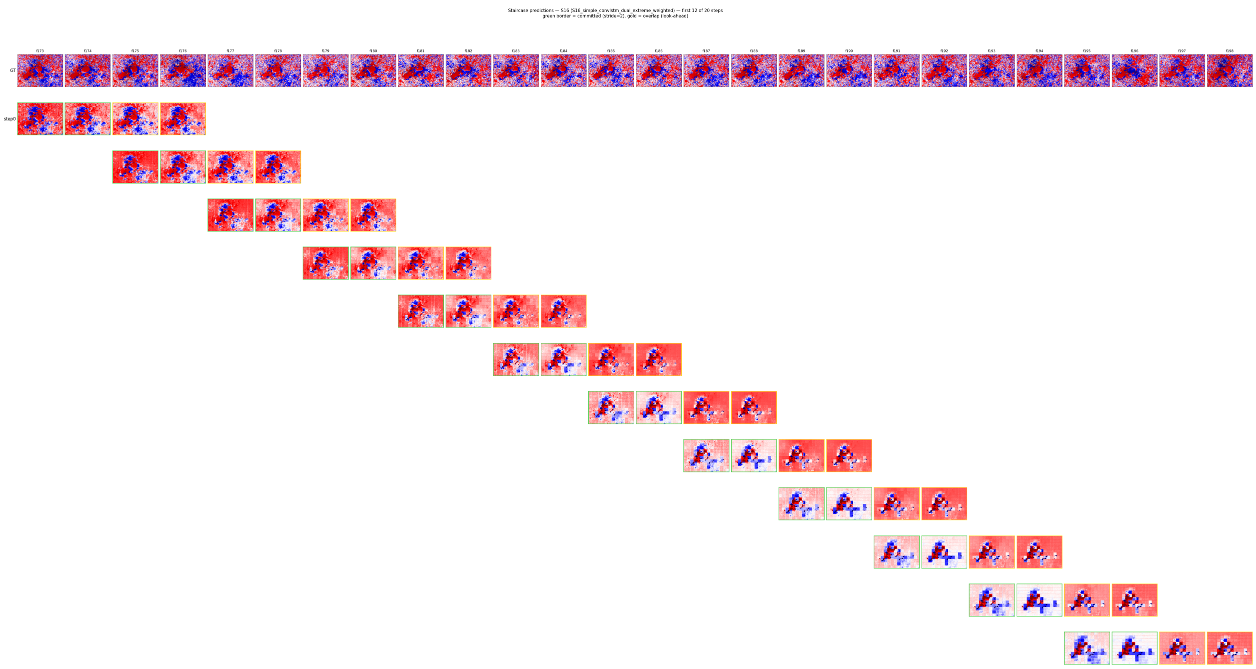




S16: staircase







Verdict

None of S3 / S11 sustains chained prediction — one collapses, the other explodes. The two failure modes are symmetric: L1-dominated losses converge to a stable low-amplitude fixed point, while amplitude-rewarding losses (S11's extreme weight + fast-TF) shift the autoregressive map to an unstable regime. Long-horizon forecasting is genuinely open; the single-step gains from S16 should be re-tested under chained rollouts before claiming multi-hour skill.

7 Cross-arm summary & next levers

Test-set CSI ranked (fixed test cubes 245, 274, 49)

Arm	reg. CSI	CLS-CSI	cls HSS	vs. pers. 0.043
tabhdr**S16** (hybrid wMAE + classifier)	**0.0255**	**0.0562**	**0.0995**	**+31% above**
S15 (α=0.1 dual)	0.0184	0.0445	0.0783	matches
tabhdrS13 (dual, p=100)	0.0090	0.0434	0.0749	matches
S11 (regression only, fastTF + ext. wt.)	0.0176	—	—	41% of
S6 (extreme weight only)	0.0215 (val)	—	—	—
S3 (full composite)	0.0236 (val)	—	—	—
S9 (BatchNorm)	0.0118	—	—	27% of
S7 (signed asinh)	0.0042	—	—	10% of
S8 (L1 fixed test)	0.0033	—	—	8% of
S10 (fast-TF L1)	0.0027	—	—	6% of
S0 (baseline)	0	—	—	0%
S14 (focal classifier)	0 reg, **0 cls**	0	0	dead arm

Mechanism narrative (one sentence per arm cluster)

- **S0–S11 (regression-only)**: L1 minimises the conditional median of a heavy-tailed field; predictions never cross the extreme threshold; CSI capped well below persistence.
- **S13 (dual-head, pos_w 100)**: explicit BCE classifier bypasses the regression bottleneck; matches persistence at the cost of a dead regression head.
- **S14 (focal)**: focal's $(1-p)^\gamma$ factor over-suppresses gradients on rare positives at our imbalance level ⇒ classifier never escapes init.
- **S15 (α=0.1)**: lowering regression weight does not buy classifier headroom; classifier saturates above some β-threshold.
- **S16 (joint hybrid)**: weighted-MAE on regression head + lighter pos_w 60 BCE classifier ⇒ both heads learn complementary signals; first arm above persistence.

Next levers

If we want to push past S16's +31%-of-persistence:

1. `pos_weight` sweep at fixed `extreme_pixel_weight`=25: find the Pareto frontier between regression and classifier heads.
2. **Quantile regression head** (pinball loss at $\tau \in \{0.5, 0.9, 0.99\}$): predict the 99-th percentile amplitude directly rather than gating an L1 mean. Skips the classifier detour.
3. **Event-onset reframe**: per-frame $P(\exists\,\text{extreme pixel in next K frames})$ using GOES-aligned event labels — orthogonal to per-pixel rate; uses the half of HARP cubes (e.g. 11930) where pixel rate and GOES events are decoupled.
4. **GradNorm balancing**: auto-balance α, β so per-batch gradient norms are equal; avoids hand tuning the multi-task loss weights.
5. **Autoregressive regularisation**: penalise variance ratio →1 during training, addressing both S3 collapse and S11 explosion identified in the staircase diagnostic (§6).

Reproducibility pointers

Configs: configs/ablations/S{0..16}_*.yaml. Best checkpoints: outputs/ablations/<arm>/checkpoints/b
Compare & staircase scripts: scripts/s0_viz/{_compare_harp11930, _s17_combo, _staircase_harp11930, _staircase_viz}.py. Per-epoch logs: logs/main__5060ti_cuda.log and logs/main__mini_mps.log.

Full per-experiment writeup in docs/V4/findings_dual_head.md (dual-head pivot), docs/V4/findings_simple_
(early regression arms), and the ablation matrix table in docs/V4/04_experiments_and_state.md.